Fiscal Year:	FY 2016	**Task Last Updated:**	FY 11/21/2015
PI Name:	Sadek, Hesham M.D., Ph.D.		
Project Title:	Long-Term Effect of Space Radiation on Cardiomyocyte Turnover		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Dallas	**State:**	TX
Zip Code:	75235-5303	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-RADIATION
Start Date:	01/20/2015	**End Date:**	01/19/2017
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Asaithamby, Aroumougame Ph.D. (University of Texas Southwestern Medical Center at Dallas)		
Grant/Contract No.:	NNX15AE06G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The risks to astronauts from exposure to space radiation have long been of great concern to NASA. These risks are now more important with the advent of long-duration space missions, which raise the risk of exposure to space radiation. While the effect of space radiation on cancer and coronary artery disease has been studied, little is known about the effect of space radiation on the heart muscle, in particular on the ability of the heart muscle to replenish itself. The adult heart is incapable of regenerating itself after injury because the vast majority of heart muscle cells cannot divide. Nevertheless, the adult human heart slowly replenishes 1-2% of its muscle every year, and it is estimated that in a human heart close to 50% of the heart muscle is renewed during its lifespan. Therefore, preventing the formation of new heart muscle cells could result in a loss of up to 50% of heart muscle leading to heart failure. We recently discovered that after birth, the increase in oxygen in the atmosphere results in formation of harmful reactive oxygen species (ROS), which in turn damage the DNA of heart muscle cells (Cardiomyocytes) and cause them to stop dividing. Intriguingly, a small population of dividing heart muscle cells is protected from DNA damage by maintaining themselves in a low oxygen environment within the heart. However, if prolonged exposure to space radiation damages the DNA of these cells, then the heart will lose any ability to replenish itself. Our proposal will determine the long-term effect of space radiation on the heart by shutting down the slow, but crucial, ability of the heart to replenish itself. The long-term goal of this project is to develop countermeasures to reduce heart failure risk due to space radiation in Astronauts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research would also lead to the development of countermeasures to enhance astronaut’s ability to sustain HZE (high energy ions) particles exposures in space.
Task Progress:	Overall, we have established that ionizing radiation has adverse effect on cycling cardiomyocytes in the adult heart and this might impact on the cardiovascular functions later in life. 1. Low-LET (linear energy transfer) ionizing radiation induced DNA double-strand breaks are irreparable in adult mouse cardiomyocytes. 2. Ionizing radiation activates DNA damage response signaling in adult heart. 3. Radiation exposure attenuates cycling cardiomyocyte turnover in adult heart. 4. Cycling cardiomyocytes are susceptible to radiation induced DSBs (double strand breaks) and elicit DNA damage response signaling.
Bibliography Type:	Description: (Last Updated: 11/12/2020)
Abstracts for Journals and Proceedings	Asaithamby A, Sadek HA. "Effect of space radiation on cardiomyocyte turnover." 2015 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 13-15, 2015. , Jan-2015